Process-Based Modelling of Greenhouse Gas Emissions in Paddy Fields Using the DNDC Model

Internship Project Report

Submitted

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Chanchal Tiwari

Abstract

Rice cultivation is a significant source of greenhouse gas (GHG) emissions, especially methane (CH_4) and nitrous oxide (N_2O), contributing to global warming. This study used the DNDC-95 model to examine the effects of different rates of biochar application on GHG emissions, soil organic carbon (SOC), and global warming potential (GWP) under Indian paddy field conditions. Separate simulations were conducted for kharif and rabi seasons, along with annual analysis. Results showed that biochar application effectively reduced CH_4 emissions during the kharif season due to improved soil aeration under flooded conditions. In the rabi season, the reductions were smaller but consistent, mainly improving SOC and slightly reducing N_2O emissions. The study found that 30-50% biochar application gave significant benefits, while higher rates showed limited additional reduction in GWP.

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Introduction

Global climate change, driven by the greenhouse effect, has become a major environmental concern, drawing significant attention from the Global community. Agriculture is one of the primary sources of greenhouse gas (GHG) emissions, accounting for approximately 58.8% of total anthropogenic GHG emissions. Within agricultural systems, paddy fields contribute about 12% of global carbon dioxide (CO_2) and 18% of methane (CH_4) emissions, making up roughly 30% of total anthropogenic carbon emissions. Collectively, CO_2 and CH_4 contribute approximately 82% to global warming potential (GWP) (Sun, 2024).

In South Asia, including India, rice production has increased significantly due to the adoption of modern agricultural technologies, such as early-maturing, nitrogen-responsive semi-dwarf cultivars, intensive use of inorganic fertilizers (particularly nitrogen fertilizers), pesticides, and the expansion of irrigation infrastructure (Pathak, 2005). However, the excessive application of nitrogen fertilizers has led to substantial nitrogen losses into the atmosphere and water bodies. Nitrous oxide (N_2O) emissions, produced through nitrification-denitrification processes, are a key pathway for nitrogen loss from paddy fields. Over a 100-year timeframe, the GWP of N_2O is 265 times greater than that of CO_2. Additionally, ammonia (NH_3) volatilization accounts for 10-50% of nitrogen fertilizer losses annually in global rice production systems. To address these issues, several studies have demonstrated that biochar (BC) amendments can effectively mitigate GHG emissions from agricultural soils (Sun, 2024).

Due to its inherent stability, biochar can enhance soil carbon sequestration by utilizing by-products of organic matter decomposition, thereby indirectly reducing CO_2 and CH_4 emissions. Its large surface area and abundant oxygen-containing functional groups contribute to controlling nitrogen emissions. Specifically, biochar can adsorb and retain NH_4^+-N, preserving soil nitrogen and reducing NH_3 volatilization at the source.

Furthermore, biochar improves soil fertility and water retention, promotes plant root growth, increases root exudate secretion, and enhances carbon inputs to the soil system (Sun, 2024).

Biochar is increasingly recognized for its potential to reduce greenhouse gas (GHG) emissions and enhance soil health, yet important questions remain. This study seeks to address**:** How does incremental biochar addition influence simulated GHG emissions over a single growing season? At what application rate do diminishing returns in mitigation begin?? And what practical recommendations can guide future research and applications?

Methodology

The Denitrification-Decomposition (DNDC) model is a process-oriented computer simulation model of carbon and nitrogen biogeochemistry in agroecosystems. The model consists of two components. The first component, consisting of the soil climate, crop growth and decomposition sub-models, predicts soil temperature, moisture, pH, redox potential (Eh) and substrate concentration profiles driven by ecological drivers (e.g., climate, soil, vegetation and anthropogenic activity). The second component, consisting of the nitrification, denitrification and fermentation sub-models, predicts emissions of carbon dioxide (CO2), methane (CH4), ammonia (NH3), nitric oxide (NO), nitrous oxide (N2O) and dinitrogen (N2) from the plant-soil systems.

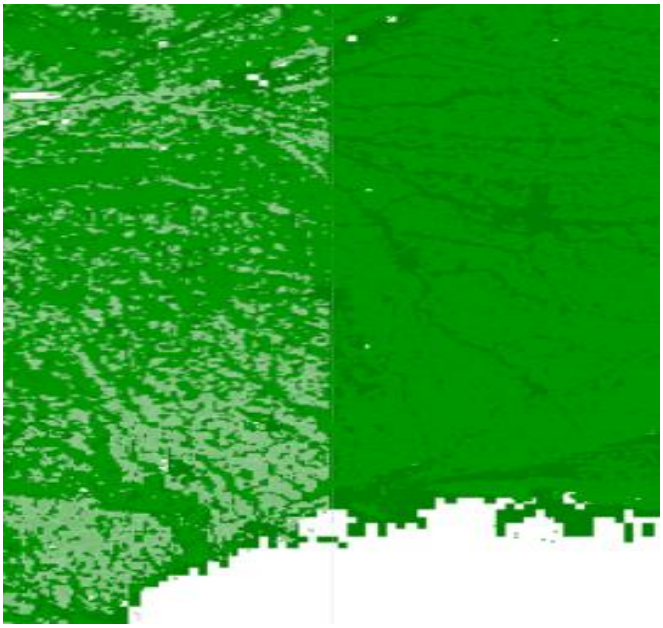
The Model works on two modes: (a) SITE Mode (b) REGIONAL Mode

The study focuses on SITE MODE of this Model.

Study Area :

The Cauvery River delta region (10°94', 78°42') in the Thiruvarur district of Tamil Nadu, Geographically it lies between 79°15' to 79°45' East longitudes and 10°20' to 11°07' North latitudes with an altitude of 10 m.

The Cauvery delta region is specifically known for paddy cultivation in the Tamil Nadu state of India in the two following seasons (Kharif or Kuruvai and Rabi or Thaladi) every year.



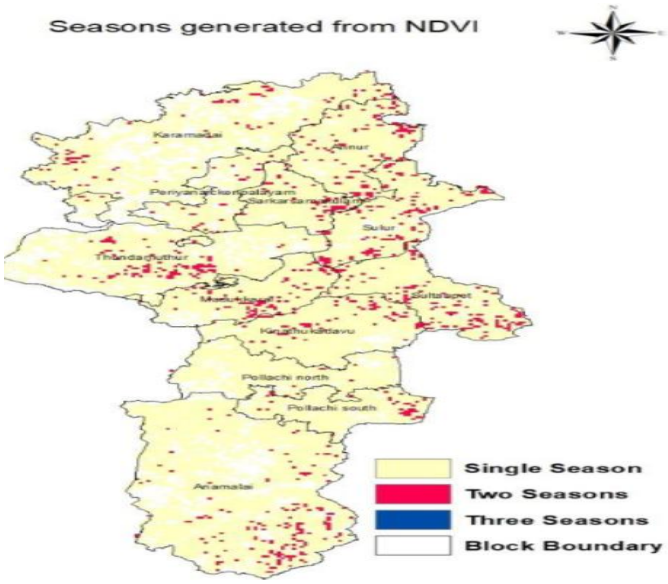


Fig.1. GEE Image of the Study Area and Source((*An Automated Tool for Extraction of Crop Condition from Temporal Synthetic Aperture Radar (SAR) Data*, n.d.)

For reference : C11.xlsx

Data Collection & Processing

(a) Climate file for site mode simulation is in the format Julian day (from 1st January 2024 to 31st December 2024), maximum temperature at 2m , minimum temperature at 2m ,and precipitation. The Climate File is in the form .*txt file and data is driven by NASA Power site (*NASA POWER*, n.d.), simulated for a year and at a latitude of 10°94'.

(b) Soil-Input for Cauvery Delta Region for all Biochar percentage:

Soil Parameters Setup	
Soil Parameter	Value Unit
Land-use type	Rice paddy field
Soil texture	loam soil
Bulk density	1.1411cm^(-3)
Soil pH	7.5
Field capacity	0.49
Wilting point	0.22
Clay fraction	0.19
Hydrological conductivity	0.025 mh^(-1)
SOC content	0.007 kg^(-1) soil
SOC partitioning(fraction of resistant litter, humads, humus and biochar)	0.01,0.0208,0.9692,0.0

The data has been taken from a research paper that is not based on Cauvery delta region because there are studies focused mainly on regional mode of this model for Cauvery region, to run for site mode, calibrated data has been used(Shaukat, n.d.).

(c) Management Practices (Separated by seasons) :

The data which has been shown in this section is not from literature but on experimental and presumed basis.

For the Kharif season, a single tillage operation was performed using the Disk/Chisel method at a depth of 10 cm in June prior to transplanting. For the Rabi season, a single tillage operation was conducted on October 1st at the same depth (10 cm) to prepare the field.

Fertilization for both seasons involved split applications of urea at a depth of 5 cm. In Kharif, applications were scheduled on June 1st, July 5th, and August 10th. For the Rabi season, urea was applied on October 15th, November 15th, and December 15th. No manure amendments were incorporated during either season.

Flooding practices followed typical paddy cultivation schedules. For the Kharif season, flooding was maintained from June 15th to September 30th. For the Rabi season, flooding occurred from October 15th to January 31st, aligning with the crop calendar for the region. No film mulch or grazing/cutting management was applied in either season.

These management practices were consistently maintained across all biochar application rates.

(d) Now, Input files have been saved annually and seasonally, and model is ready to run and for desirable results.

Result

R.1

Annual cumulative results confirm the consistent benefit of biochar across both seasons. Reductions in GWP were gradual but more significant when observed cumulatively over the year.

Biochar(%)	Change in Soil C storage(Kg C/ha/yr)	N_2O Flux(Kg N/ha)/GWP	CH_4 Flux(Kg C/ha)/GWP	Net Global Warming Potential(Kg CO_2-equivalent/ha)	% Reduction in Net GWP from Control
0	3876.1	2.3/1125	178/4980	-8107	0.00%
1	3877.8	2.3/1120	178/4979	-8120	0.16%
2	3879.5	2.3/1119	178/4977	-8127	0.25%
3	3881.3	2.3/1116	178/4976	-8140	0.41%
4	3883	2.3/1116	178/4974	-8148	0.50%
5	3884.7	2.3/1111	178/4972	-8161	0.67%
10	3893.1	2.3/1096	177/4964	-8215	1.33%
30	3927.5	2.1/1042	176/4930	-8429	3.96%
50	3963	2.0/994	175/4892	-8646	6.65%
70	3998.9	2..0/950	173/4845	-8867	9.36%
90	4048.1	1.8/882	171/4780	-9181	13.27%
96	4048.1	1.8/882	171/4780	-9181	13.27%

Negative value means the system is acting as a net sink for greenhouse gases that is it's removing more CO_2-equivalent than it emits (good for the climate).

Whereas Positive value would mean a net source that is it emits more CO_2-equivalent than it sequesters (bad for the climate).

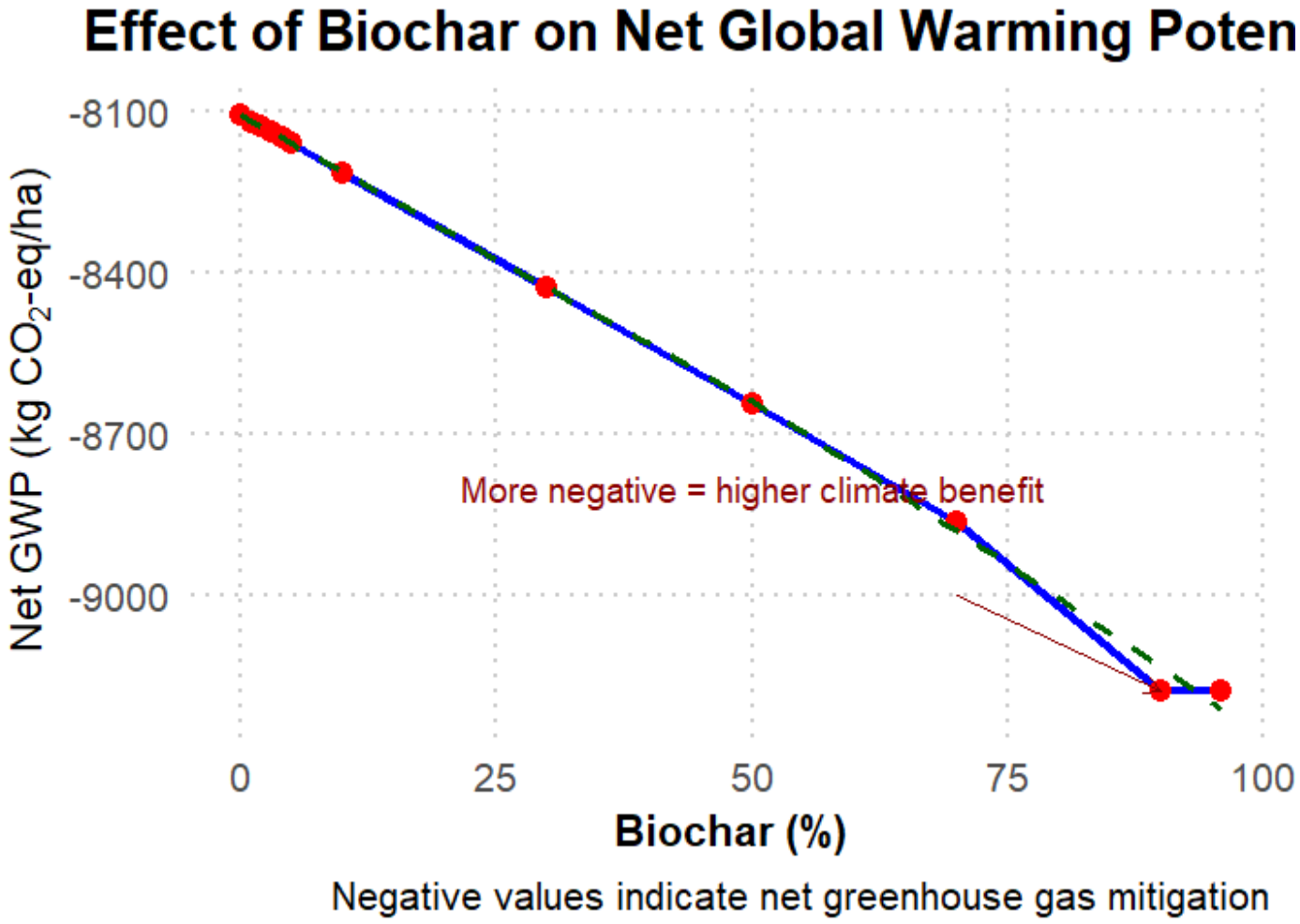


Fig.2.

Observation: Up to 13.27% annual reduction in net GWP with 90-96% biochar.

R.2

The relationship between biochar application rates and net GWP during the Kharif season showed a consistent decrease in net GWP with increasing biochar percentage. Methane (CH_4) emissions dominated the GWP during this flooded season, while nitrous oxide (N_2O) emissions remained relatively constant.

Biochar(%)	Change in Soil C storage(Kg C/ha/yr)	N_2O Flux(Kg N/ha)/GWP	CH_4 Flux(Kg C/ha)/GWP	Net Global Warming Potential(Kg CO_2-equivalent/ha)	% Reduction in Net GWP from Control(Mod)
0	-175.2	8.1/3941	444/12418	17002	0.00%
1	-174	8.1/3936	443/12392	16966	0.21%
2	-172.8	8.1/3936	442/12376	16937	0.38%
3	-171.3	8.1/3937	441/12358	16922	0.47%
4	-170.1	8.1/3931	440/12330	16884	0.69%
5	-168.9	8.1/3931	439/12301	16852	0.88%
10	-163.1	8.1/3923	434/12164	16683	1.88%
30	-138	8.0/3878	417/11689	16072	5.47%
50	-119.2	7.9/3829	390/10910	15177	10.75%
70	-89.2	7.8/3785	380/10632	14744	13.27%
90	-61.1	7..7/3741	367/10273	14238	16.34%
96	-52.8	7.7/3727	363/10156	14077	17.18%

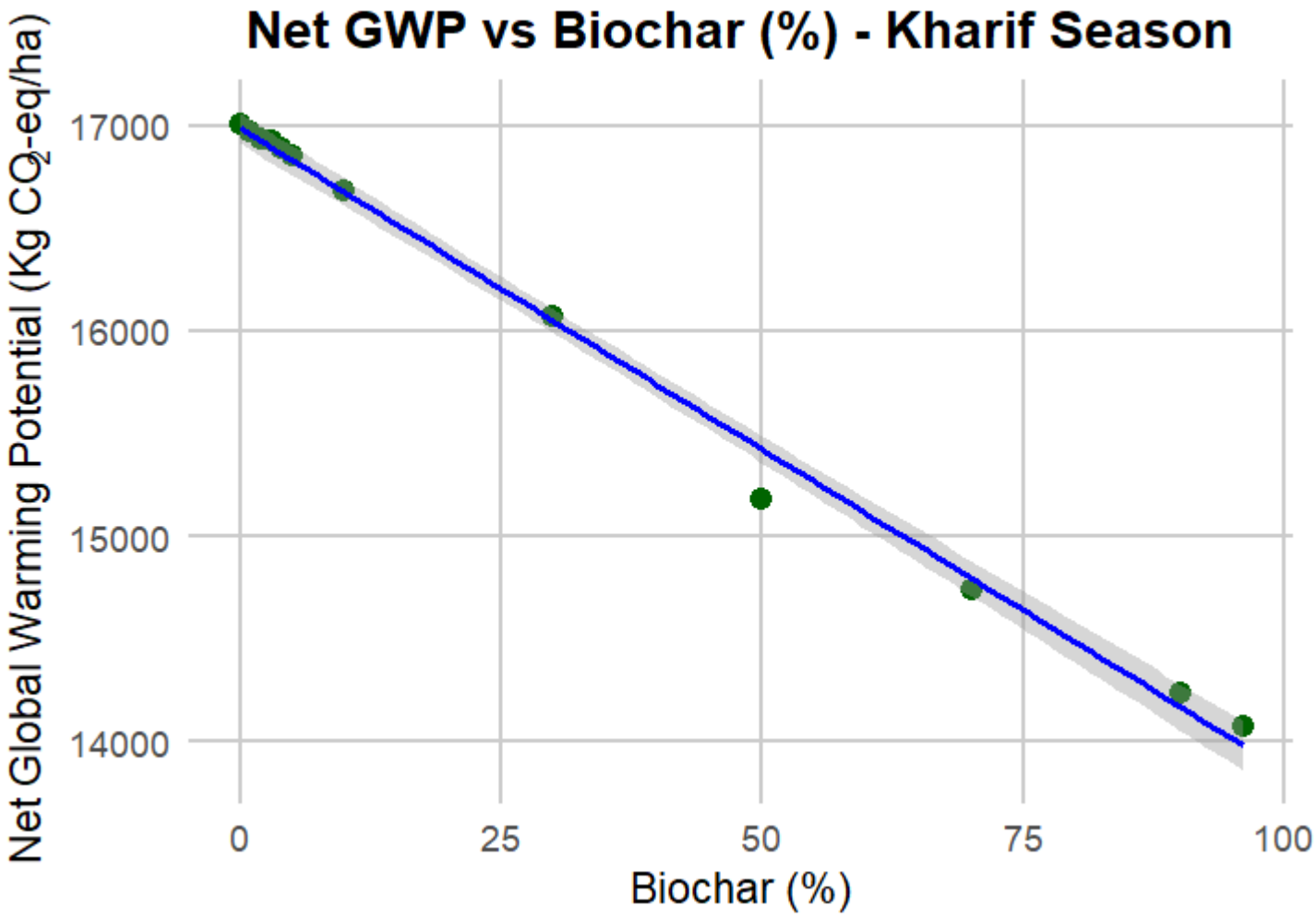


Fig.3.

Observation: A clear downward trend, reduction up to 17.18% at maximum biochar rate.

Accuracy: It's R squared, RMSE and MAE values are 0.9945989, 81.27516 and 49.8784

R.3

In the Rabi season, biochar application also resulted in decreasing net GWP, though reductions were more moderate than Kharif. This reflects the less flooded conditions and reduced CH_4 emissions, with N_2O emissions remaining steady.

Biochar(%)	Change in Soil C storage(Kg C/ha/yr)	N_2O Flux(Kg N/ha)/GWP	CH_4 Flux(Kg C/ha)/GWP	Net Global Warming Potential(Kg CO_2-equivalent/ha)	% Reduction in Net GWP from Control(Mod)
0	208.6	0.4/180	237/6644	6060	0.00%
1	209.9	0.4/181	237/6635	6046	0.23%
2	211.3	0.4/182	237/6626	6031	0.48%
3	213.8	0.4/183	236/6621	6017	0.71%
4	215.1	0.4/175	236/6611	5998	1.02%
5	216.3	0.4/176	236/6600	5983	1.27%
10	222.6	0.3/171	234/6548	5902	2.60%
30	261.4	0.3/156	228/6391	5589	7.77%
50	273.2	0.3/146	218/6097	5242	13.49%
70	272.8	0.3/141	205/5732	4873	19.63%
90	283.6	0.3/136	191/5352	4448	26.61%
96	291.4	0.3/136	180/5200	4300	29.05%

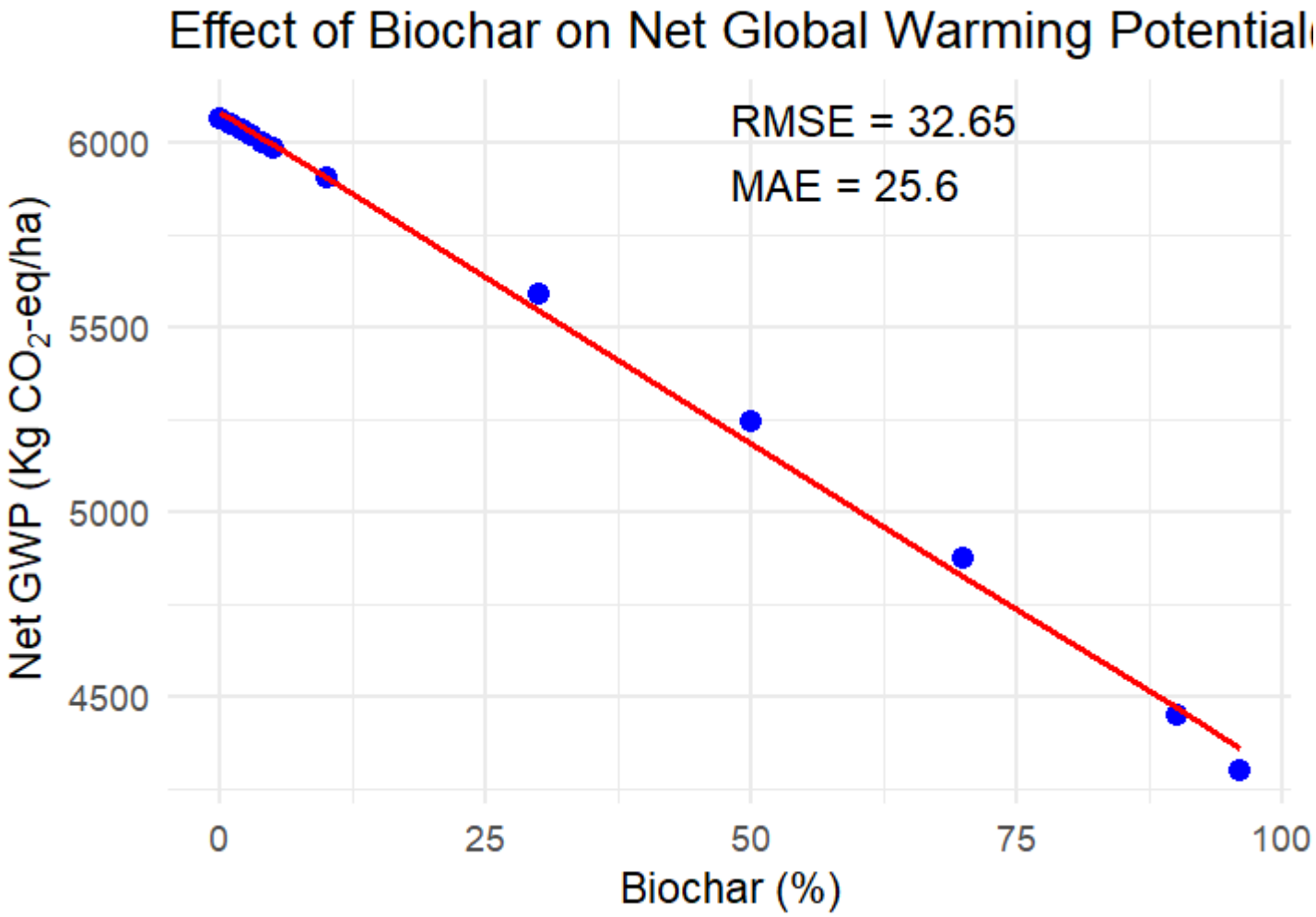


Fig.4.

Observation: Reduction reached up to 29.05% at maximum biochar rate.

Accuracy: It's R squared, RMSE and MAE values are 0.9973341, 32.64938 And 25.60281.

R.4

Greenhouse Gas Emissions (GHG) fluctuations across two seasons

- Kharif: Higher CH_4 emissions due to continuous flooding; mitigation through biochar more pronounced.
- Rabi: Lower CH_4 emissions, but biochar still beneficial through SOC enhancement and slight reductions in N_2O.
- Biochar had a stronger mitigation impact during the Kharif season (monsoon flooding).

R.5

Summary of the results

The following trends were observed when comparing biochar % (input) against modelled outputs (Net GWP reduction):

- Increasing biochar consistently reduced GWP.
- The reduction trend plateaued beyond 50% biochar application, indicating diminishing returns.
- Seasonal breakdown showed higher sensitivity in Kharif due to CH_4 dynamics.

Key Findings and model uncertainty:

- Incremental biochar consistently reduces simulated GHG emissions, especially CH_4 during Kharif.
- Diminishing returns start beyond 50% biochar application.

- DNDC-95 results align well with literature on Cauvery delta GHG dynamics.
- Practical recommendation: focus on 30-50% biochar for optimal cost-benefit in field applications.
- The DNDC-95 model does not explicitly simulate biochar aging effects or SOC stability beyond decadal scales.
- Results reflect modelled scenarios; field-scale uncertainties (weather variability, soil heterogeneity) are acknowledged.

For reference: c12.xlsx

Discussion

Seasonal trends in greenhouse gas emissions were clearly observed. In the kharif season, methane (CH_4) emissions were higher due to continuous flooding, but biochar application effectively reduced these emissions. In the rabi season, CH_4 emissions were lower because of drier field conditions, and the impact of biochar was moderate, contributing to slight reductions in nitrous oxide (N_2O) emissions and improvement in soil organic carbon (SOC).

The mitigation effect of biochar was more noticeable during the kharif season because of the flooded conditions. In the rabi season, the reductions were smaller but consistent. Across both seasons, biochar contributed positively to soil carbon storage.

The reductions in net global warming potential (GWP) began to stable after 50% biochar application, indicating diminishing returns beyond this level. Therefore, a 30-50% biochar application appears to offer an effective balance between environmental benefits and resource efficiency.

Conclusion

In this study, the DNDC-95 model was used to see how different amounts of biochar affect greenhouse gas emissions, soil carbon, and global warming potential in rice fields in India.

The results showed that adding more biochar helped reduce greenhouse gases, especially methane during the kharif season when the fields stay flooded. In the rabi season, the reduction was smaller but still useful, mainly because biochar helped improve the soil and reduce nitrous oxide a little.

The study found that using 30-50% biochar is a good choice. Adding more than this did not show much extra benefit.

However, the model cannot show how biochar works in the very long term. Future work should improve this and look at combining biochar with other farming methods like saving water through better irrigation, say Controlled Irrigation (CI)

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